

QRM II Graded Assignment (3)

Period 1, 2026

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0 Introduction

This assignment is to be completed in groups of 4-5. Further, all students in your group need to be assigned to the same R tutorial group (Friday's tutorial). You can sign yourself up for a group on Canvas. Please do so **before the start of your first R tutorial on Friday September 4th**. You can use the Discussion Board in Canvas if you do not have a group yet or if your group is incomplete.

The assignment has 5 parts, and each part corresponds to the course material of that week (with the exclusion of week 6, for which there is no R programming material).

You are supposed to hand in these assignments on Canvas at the following dates:

- **Deadline 1** *Sunday September 27th, at 23:59pm*: you are supposed to hand in weeks 1, 2, and 3 of this assignment. This will determine 18% of your overall course grade
- **Deadline 2** *Sunday October 11th, at 23:59pm*: you are supposed to hand in weeks 4, and 5 of this assignment. This will determine 12% of your overall course grade

The R tutorials (each Friday) will consist of two halves. During the first half, you will discuss the tutorial exercises. These can be downloaded separately from Canvas. During the second half, you can work on this graded assignment within your own group. The purpose is that you find out how to work with R for doing statistical analyses by yourself. The tutorial exercises are meant to teach you basic commands to get you started, but to answer the problem sets in this assignment, you might need to research your own solutions, and use functions and commands not described in the tutorial exercises. Learning how to solve your own research problems is integral part of learning R. When you and your group get stuck on how to approach an exercise, the hierarchy in finding your way is as follows:

- use the concepts from the tutorial exercises;
- use the cheat sheets available on Canvas;
- use Google, YouTube, StackOverflow, or another website;
- ask the teacher.

The use of generative AI is **not** permitted and may result in a grade of 0. See the AI protocol in the course manual for details.

To answer the assignment, you can simply fill out this R markdown document. There are designated places which you can fill with R code. There are also designated spaces for you to answer each question. Often, the structure of an answer will be as follows. First, you type the R code in the designated box. This will show how you analyzed the data to get the answer to the question. Below the box for the R code, you will then summarize your answer to the question, i.e. what are the conclusions that you draw from the data analysis?

When handing in, you are supposed to submit this .Rmd file, and a knitted version of this document. You can knit this document to pdf, word, or html. Knitting to pdf requires you to have a .tex distribution installed on your computer. Knitting to Word requires you to have Word installed.

The exercises are designed such that you should be able to finish the majority of them during the tutorial each week. If you are not able to finish them fully during that time, you are expected to work on it in your own time using the computers on campus or your own device. It is best to meet as a group in-person when working together. If you want to work remotely, github is a good platform to guarantee smooth collaboration. Alternatively, you can email this .Rmd file back and forth to one another as a group, but this is not recommended as it is more cumbersome.

We encourage you to keep your code blocks, printing statements, and final answers, as short as possible. In any case, there is a page limit of 6 pages per week, which encompasses the total length of this document which consists of the questions, your coding lines, and your answers. When your answers to questions of the respective week exceed this page limit, they will not be graded, resulting in zero points.

1 Week 1

1. Find the dataset “movies5.tsv” on Canvas. Describe your data set: How many observations does it have. How many variables are there? How many subjects? What consists of a subject? [4 points]

```
movies <- read_tsv("Data/movies5.tsv")

## Rows: 909 Columns: 19
## -- Column specification -----
## Delimiter: "\t"
## chr   (8): keywords, original_language, title, genre, first_actor, first_act...
## dbl  (10): index, budget, popularity, revenue, runtime, vote_average, vote_c...
## date  (1): release_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

movies5 <- movies

dim(movies5)

## [1] 909 19

n_distinct(movies5$title)

## [1] 909
```

Your Answer: It has 909 observations, 19 variables and 909 subjects. One subject is equal to one film in this dataset. Every row in the dataset stands for one film and that subject is described using 19 variables. There are 909 subjects because there is no film that appears twice in the dataset.

2. Which of the following types of variables are present in your data set? (i) nominal; (ii) ordinal; (iii); continuous; (iv) discrete. If present, name one example of such a variable present in your data set. [4 points]

```
sapply(movies5, class)
```

```
##           index           budget           keywords original_language
##      "numeric"      "numeric"      "character"      "character"
##           title      popularity      release_date           revenue
##      "character"      "numeric"           "Date"      "numeric"
##           runtime      vote_average      vote_count           genre
##      "numeric"      "numeric"      "numeric"      "character"
##      release_year      release_month      release_day      first_actor
##      "numeric"      "numeric"      "numeric"      "character"
## first_actor_gender director_first_name director_gender
##      "character"      "character"      "character"
```

Your Answer:

- (i) nominal is present in our dataset, for example the genre. (iii) continuous is also present, for example the popularity of the movie. (iv) discrete is also present, vote_count is a discrete variable.
3. A movie studio wants to know which types of movies give maximal revenue. Perform the following steps to provide the movie studio with an analysis which corresponds to their request:
 - a. Summarize the variable revenue and report its mean, median, maximum, and minimum. **[2 points]**
 - b. Which movie has the highest revenue in your data set and how much is this revenue. Which movie has the lowest and how much is its revenue? If multiple movies have the exact same highest or lowest revenue, give only one example. **[2 points]**
 - c. Create a histogram of the variable revenue. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Describe the shape of the histogram. What does this tell you about the nature of making money in the movies industry? **[2 points]**
 - d. Add a new variable to your data set: the log of revenue. When creating this variable, what happens to movies for which revenue is zero? What then happens when you calculate the mean of log revenue? **[2 points]**
 - e. For movies that have a revenue of zero, replace log of revenue with “NA”. What is now the mean of log of revenue? Create a histogram for log revenue, again with an appropriate title, x- and y-axis labels. Describe the shape of this histogram. **[2 points]**
 - f. For the variables revenue and log of revenue calculate their skewness and kurtosis. How do these numbers relate to the shape of the histograms that you estimated under d.) and e.)? **[2 points]**

For each step, you should provide first all the code you used to answer the question and then formulate an answer using full sentences.

Each week consists of 1, 2, or 3 subquestions. The total amount of points you can earn per week is 20 points.

Step a

```
mean(movies5$revenue)
```

```
## [1] 88846315
```

```
median(movies5$revenue)
```

```
## [1] 25482931
```

```
max(movies5$revenue)
```

```
## [1] 1156730962
```

```
min(movies5$revenue)
```

```
## [1] 0
```

Your Answer:

The mean of the revenue in US dollars is 88,846,315, the median in US dollars is 25,482,931, the maximum in US dollars is 1,156,730,962 and the minimum is 0.

Step b

```
movies5$title[which.max(movies5$revenue)]
```

```
## [1] "Minions"
```

```
max(movies5$revenue)
```

```
## [1] 1156730962
```

```
movies5$title[which.min(movies5$revenue)]
```

```
## [1] "The Brothers McMullen"
```

```
min(movies5$revenue)
```

```
## [1] 0
```

```
sum(movies5$revenue == 0)
```

```
## [1] 257
```

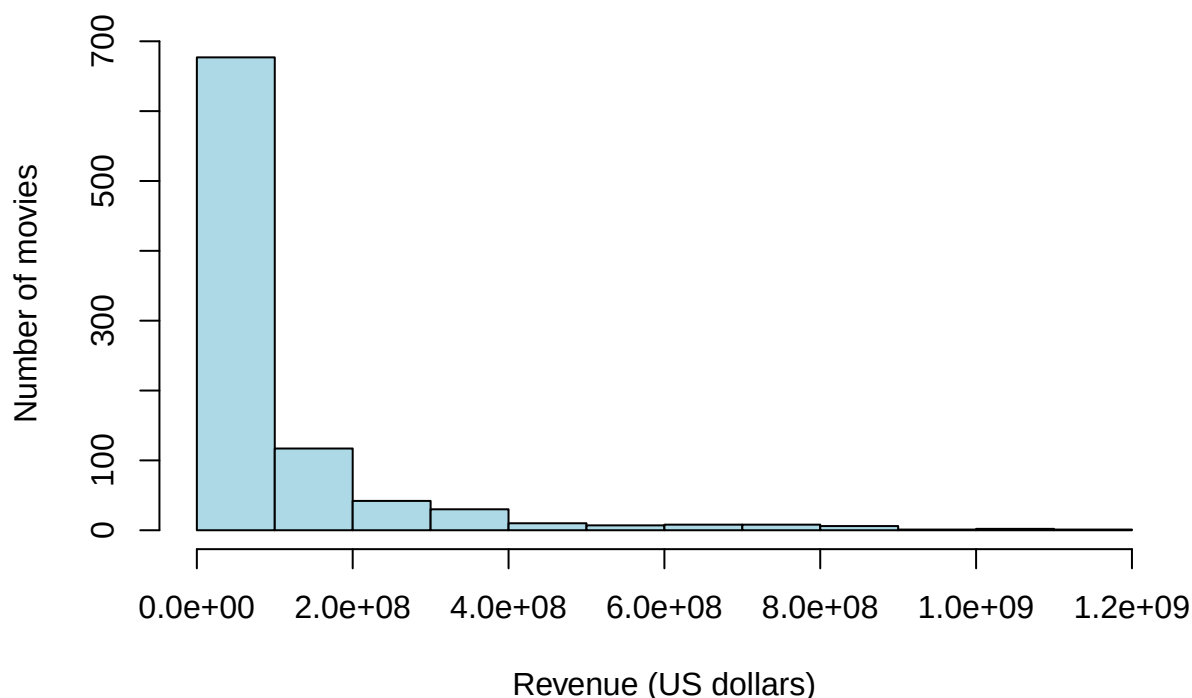
Your Answer:

The movie with the highest revenue is 'Minions' and it has a revenue of 1156730962. The movie with the lowest revenue is 'The Brothers McMullen' and this movie has no revenue, it is equal to zero.

Step c

```
hist(movies5$revenue,  
     main = "Distribution of movie revenue",  
     xlab = "Revenue (US dollars)",  
     ylab = "Number of movies",  
     col = "lightblue")
```

Distribution of movie revenue



Your Answer: The shape of the histogram: It is right-skewed with a long right tail. The Mean is far above the median because most movies perform under the mean and only 2.5 percent of the revenue belongs to the lower half.

What this tells us about the nature of making movies is that it is risky, unpredictable and only a number of movies carry the whole industry. 91 films earn 55 percent of all revenue.

Step d

```
movies5$log_revenue <- log(movies5$revenue)
sum(is.infinite(movies5$log_revenue))
```

```
## [1] 257
```

```
mean(movies5$log_revenue)
```

```
## [1] -Inf
```

Your Answer:

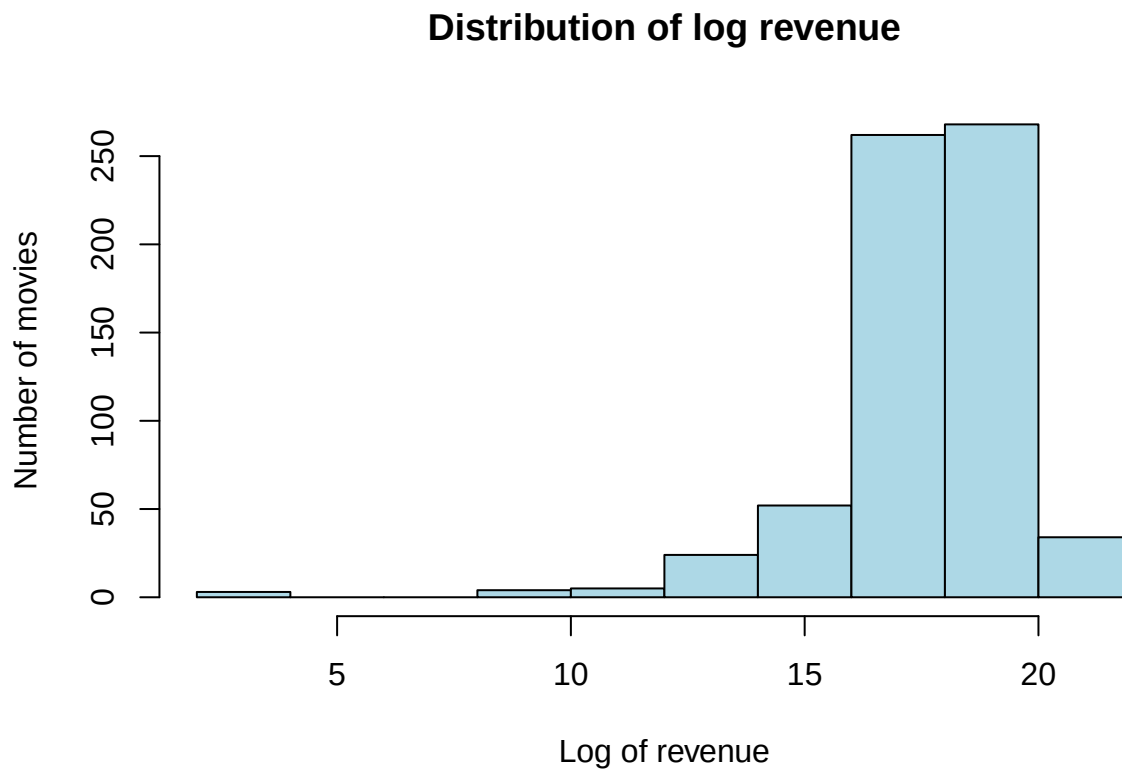
There are 257 films that have 0 revenue and their log revenue becomes -inf. The mean of the new variable is -inf, this is because when a movie has 0 revenue it is equal to $\log(0) \rightarrow -\text{Inf}$. Therefore when there is one -inf in the sum for calculating the mean it is still -inf.

Step e

```
movies5$log_revenue[movies5$revenue == 0] <- NA
mean(movies5$log_revenue, na.rm = TRUE)
```

```
## [1] 17.54381
```

```
hist(movies5$log_revenue,
     main = "Distribution of log revenue",
     xlab = "Log of revenue",
     ylab = "Number of movies",
     col = "lightblue")
```



Your Answer: The mean of log of revenue 17.54381. The mean only covers films with revenue above 0. The shape of this histogram is left-skewed, tallest bars are on the right of the x-axis between 16 and 20 and a long thin tail runs to the left to about 2.

Step f

```
skewness(movies5$revenue)
```

```
## [1] 3.181042
```

```
kurtosis(movies5$revenue)
```

```
## [1] 14.82708
```

```
skewness(movies5$log_revenue, na.rm = TRUE)
```

```
## [1] -2.514868
```

```
kurtosis(movies5$log_revenue, na.rm = TRUE)
```

```
## [1] 15.41509
```

Your Answer:

For the revenue variable the skewness is positive and is 3.181042 which matches the right skewed histogram of step c. The log revenue skewness is -2.514868 and is negative which matches the left skewed histogram of e. The revenue kurtosis is 14.82708 and the log revenue kurtosis is 15.41509. Both have kurtosis of around 15 which is far above the 3 which is the reference point for normal distribution which explains the graph of both. What kurtosis far above 3 means: more extreme values in the tails than a normal distribution.

2 Week 2

1 Is your dataset movies5.tsv the full population, or is it a sample of a larger population? If the latter, how would you describe the full population? **[4 points]**

Your answer:

it is a sample from a larger population. there have been arguably hundreds of thousands of movies created throughout history. 909 is too little to represent every movie ever made. a good description of the full population would be “English language feature movies from 1990-2016 in the TMDb database”.

2 For this question, you will assume that your data set is the full population.

- What is the mean of the variable vote_average in your data set? **[2 points]**
- Create a new data set, called movies_sample. Make sure that it is a random sample of your data set of 25 movies. What is the mean of vote_average in this random sample? How does it compare to the mean of vote_average in 2a? **[2 points]**
- In a for loop, create 100 different samples of 25 movies, as in b, and estimate the mean within each sample. Save the mean of each sample in a vector called sample_means. So the first position of the vector would have the mean of the first sample, the second position the mean of the second sample, etc. Print the start of this vector. **[2 points]**
- Summarize and make a histogram of sample_means. What is the mean, standard deviation and shape of its distribution? **[2 points]**
- Using the formulas of the *Central Limit Theorem*, give the distribution of the sample mean of vote_average in a sample of 25 movies from your population (including its expected value and standard deviation). How do these values compare to your answers at d.)? **[2 points]**

Step a

```
mean(movies5$vote_average)
```

```
## [1] 6.011001
```

Your answer:

The mean of vote_average across the full data set is 6.011001.

Step b

```
set.seed(123)
movies_sample <- movies5[sample(nrow(movies5), 25), ]
mean(movies_sample$vote_average)
```

```
## [1] 6.08
```

Your answer:

using a reproducible random seed, the result is 6.08. this value is slightly higher than the population mean, due to natural sampling variability. A little sample of 25 movies won't reflect the whole population.

Step c

```
sample_means <- numeric(100)
for (i in 1:100) {
  s <- movies5[sample(nrow(movies5), 25), ]
  sample_means[i] <- mean(s$vote_average)
}
head(sample_means)
```

```
## [1] 6.232 5.852 5.848 5.856 5.748 6.236
```

Your answer :

These are the first 6 of the 100 sample means. the vector stores the estimated vote_average means for 100 uniquely drawn samples of 25 movies for each one.

Step d

```
summary(sample_means)
```

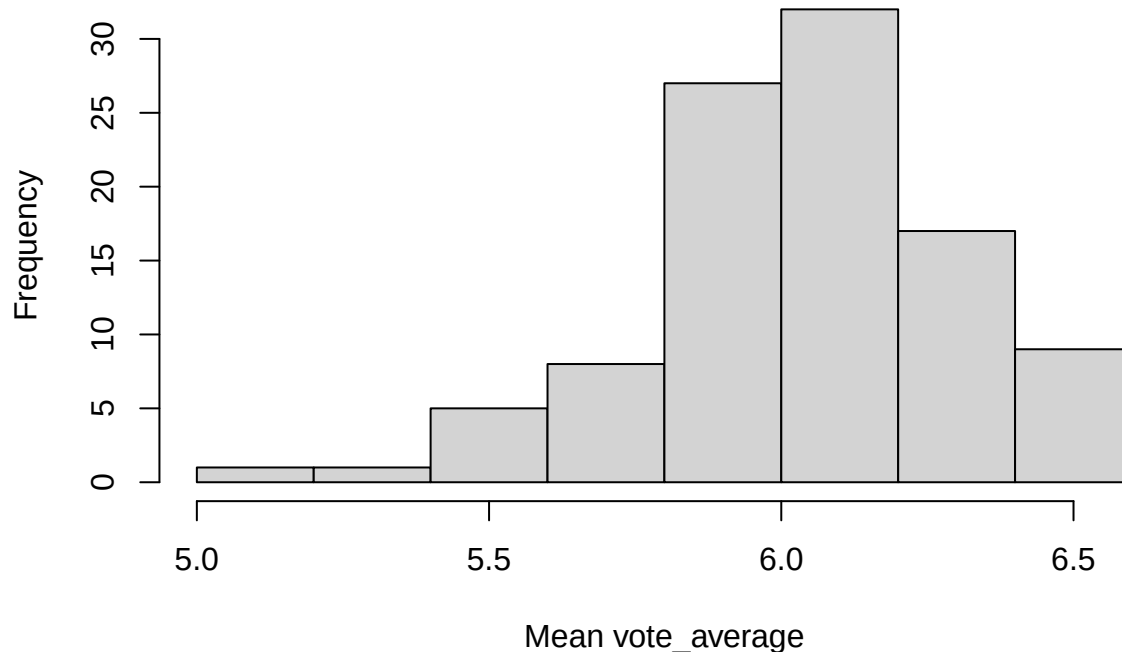
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   5.168   5.913   6.032   6.034   6.217   6.516
```

```
sd(sample_means)
```

```
## [1] 0.2648297
```

```
hist(sample_means, main = "Distribution of sample means", xlab = "Mean vote_average")
```

Distribution of sample means



Your answer:

The mean of sample_means (6.034) is nearly equal to the population mean(6.011). the standard deviation (0.265) is very close to the theoretical standard error (pop. sd = $1.226 / 5 = 0.245$). the shape is roughly symmetric and bell shaped, which is in line with the CLT.

Step e

Your answer :

The central limit theorem explains that in the situation of a sample size $n=25$, the sample mean follows $\bar{X} \sim N(\mu = 6.011, 1.226 / 5 = 0.245)$. the expected value is the population mean of vote_average, which is 6.011. the standard deviation of the sampling distribution is 0.245. so the two sets of these values are very close to each other. this is logical since the CLT formula tells exactly what sampling distribution of the mean should look like, while the simulation in question d with 100 samples gives an empirical approximation of the same distribution. the very small difference between the values is due to the finite number of simulated numbers, instead of an infinite number of samples. when using more simulations the mean and standard deviation would get even closer to the theoretical values (6.034 vs 6.011 and 0.265 vs. 0.245).

- Is there a variable in your data set which you would expect to follow a discrete uniform distribution? Explain, without looking at the data, why you expect this data to be discrete uniform. If the data would indeed follow a discrete uniform distribution, what would you expect the mean and standard deviation of this variable to be? Next, plot a histogram of the variable, and plot an appropriate discrete uniform distribution on top of this histogram. Comment whether the variable indeed follows this expected distribution. If the distribution deviates from the discrete uniform distribution, explain why this is the case. [6 points]

Your answer: expected mean= $(1+31)/2 = 16$ standard deviation= $\sqrt{((31^2 - 1)/12)} = \sqrt{80}$ a good pick would be release_day. studios prefer certain days for release, but the weekdays are spread evenly over

days on a month, so there is no reason to prefer a specific day. release day should be in principle random with respect to which day in specific it falls on. so there should be discrete uniform distribution. release_month would not work, due to seasonal peaks.

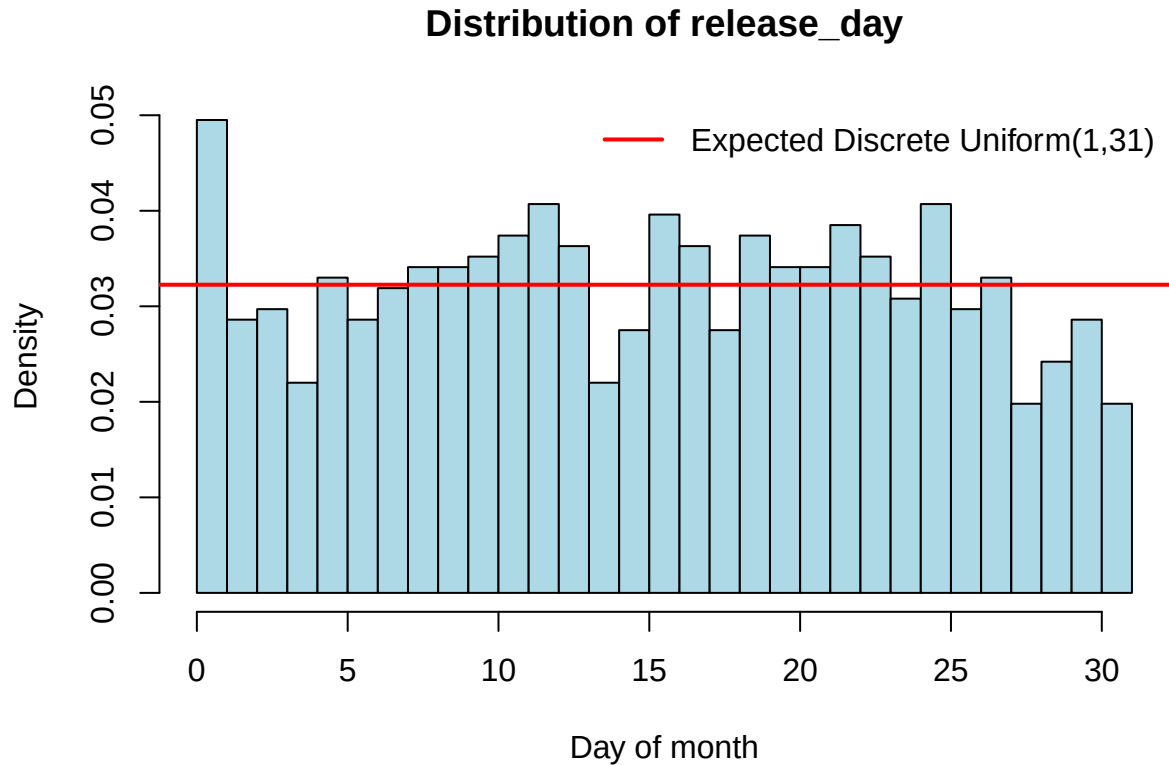
```
release_day <- movies5$release_day

# theoretical values
n <- 31
theo_mean <- (1 + n) / 2
theo_sd <- sqrt((n^2 - 1) / 12)

# observed values
obs_mean <- mean(release_day)
obs_sd <- sd(release_day)

# histogram (density scale so it's comparable to the uniform pmf)
hist(release_day, breaks = 0:31, freq = FALSE,
     main = "Distribution of release_day",
     xlab = "Day of month", col = "lightblue")

# overlay expected discrete uniform density (flat line at 1/31)
abline(h = 1/31, col = "red", lwd = 2)
legend("topright", legend = "Expected Discrete Uniform(1,31)",
     col = "red", lwd = 2, bty = "n")
```



```
theo_mean; theo_sd
```

```
## [1] 16
```

```
## [1] 8.944272
```

```
obs_mean; obs_sd
```

```
## [1] 15.52805
```

```
## [1] 8.695316
```

Your answer:

the values of the observed mean and SD again come close to the theoretical values (mean=16, SD= 8.94). the distribution is almost flat, however there are two deviations looking at the histogram. 1: there is a sharp spike at day 1. day 1 is more noticeable than any other day, which is called a data artifact. it means that when exact release dates are not known by databases, they default to january 1st, inflating the bar at day 1. 13 of the 45 day-1 movies are in january, while other months only have 1 to 5 each, close to the expected 2.5. 2: fewer counts on days 29-31. this is a structural observation since not every month has 29,30 or 31 days. higher day numbers have fewer opportunities to occur during a year, leading to a distribution which is slightly away from a perfect uniform shape.

3 Week 3

For the next part of the assignment, assume that the movies in your data frame are a random sample of a larger population of movies.

1. Give a 90 percent confidence interval for the variable *revenue*. Interpret the result. [2 points]

```
movies <- read_tsv("Data/movies5.tsv")
```

```
## Rows: 909 Columns: 19
## -- Column specification -----
## Delimiter: "\t"
## chr   (8): keywords, original_language, title, genre, first_actor, first_act...
## dbl  (10): index, budget, popularity, revenue, runtime, vote_average, vote_c...
## date  (1): release_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
t.test(movies$revenue, conf.level = 0.90)
```

```
##
## One Sample t-test
##
## data:  movies$revenue
```

```
## t = 16.666, df = 908, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 90 percent confidence interval:
## 80068897 97623733
## sample estimates:
## mean of x
## 88846315
```

Your answer $t = 16.666$, $df = 908$, $p\text{-value} < 2.2e-16$ alternative hypothesis: true mean is not equal to 0
90 percent confidence interval: 80068897 97623733 sample estimates: mean of x 88846315

We are 90% confident that the true population mean revenue of movies lies between \$80,068,897 and \$97,623,733 (approximately \$80.07 million and \$97.62 million)

2

- How many movies in your sample were directed by a male director? How many by a female director? **[2 points]**
- Test the null hypothesis that the chance that a movie is being directed by a male or female director is equal. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Use an appropriate R function to test your hypothesis, and state your conclusion. Also give the p-value that corresponds to your test statistic. **[2 points]**
- Recode revenue such that it is measured in millions of dollars. What is the variance of revenue for movies with a male director? What is the variance of revenue for movies with a female director? **[2 points]**
- Assume that the variance of revenue for movies with a male director in your data is the same as the variance for movies with a male director in your population (that is, there is no sampling uncertainty in this estimate). Now, set up a hypothesis test to test for the null hypothesis that the variance of revenue for movies with a female director is equal to the variance for movies with a male director. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Calculate your test statistic and tell how your test statistic is distributed, and state your conclusion. Also give the p-value that corresponds to your test statistic. **[2 points]**

step a

```
movies %>% count(director_gender)
```

```
## # A tibble: 3 x 2
##   director_gender     n
##   <chr>           <int>
## 1 female             82
## 2 male             780
## 3 <NA>              47
```

Your answer

In our sample, 780 movies were directed by a male director and 82 movies were directed by a female director. (for 47 movies, the director's gender was missing/NA)

step b

```
binom.test(c(780, 82), p=0.5)
```

```
##
## Exact binomial test
##
## data: c(780, 862)
## number of successes = 780, number of trials = 862, p-value < 2.2e-16
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
##  0.8832985 0.9236273
## sample estimates:
## probability of success
##          0.9048724
```

Your answer

Null hypothesis: $p=0.5$ (The probability of a movie having a male or female director is equal)

Alternative hypothesis: $p \neq 0.5$ (The probability of a movie having a male or female director is not equal).

Significance level: 0.05

RESULTS: $p < 0.001$. 95% confidence interval: [0.883, 0.924]

780 male directors out of 862 total movies Sample proportion: 0.905 (90.5%)

Since the p-value is far below the significance level of 0.05, we reject the null hypothesis. There is statistically significant evidence that the probability of a movie being directed by a male or female director is not equal. Male directors are significantly more common.

step c

```
movies$revenue_mil <- movies$revenue / 1000000
var(movies$revenue_mil[movies$director_gender == "male"], na.rm = TRUE)
```

```
## [1] 27072.45
```

```
var(movies$revenue_mil[movies$director_gender == "female"], na.rm= TRUE)
```

```
## [1] 18066.64
```

Your answer

27072.45= the variance of the revenue of films with a male director. 18066.64= the variance of the revenue of films with a female director. The variance in revenue is higher for movies directed by male directors than for those directed by female directors. This suggests that the financial performance of male-directed movies exhibits greater dispersion around the mean revenue.

step d

```
s2_female <- var(movies$revenue_mil[movies$director_gender == "female"], na.rm = TRUE)
n_female <- sum(!is.na(movies$revenue_mil[movies$director_gender == "female"]))
sigma2_0 <- 27072.45

df <- n_female - 1

chi2_stat <- (df * s2_female) / sigma2_0
```

```
p_val <- 2 * min(pchisq(chi2_stat, df), 1 - pchisq(chi2_stat, df))

chi2_stat
```

```
## [1] 54.05488
```

```
df
```

```
## [1] 81
```

```
p_val
```

```
## [1] 0.01840069
```

Your answer Test statistic: Chi-square= 54.05. Distribution: Chi-squared distribution= 81. (degrees of freedom). The p-value is 0.0184.

Null hypothesis: The variance of revenue for movies directed by female directors is equal to 27,072.45(the variance for male directors).

Alternative hypothesis: The variance of revenue for movies directed by female directors is not equal to 27,072.45.

Significance level (alpha): 0.05

Since the p-value (0.0184) is less than the significance level of 0.05, we reject the null hypothesis. There is statistically significant evidence that the variance of revenue for movies directed by female directors differs from the variance for movies directed by male directors.

3

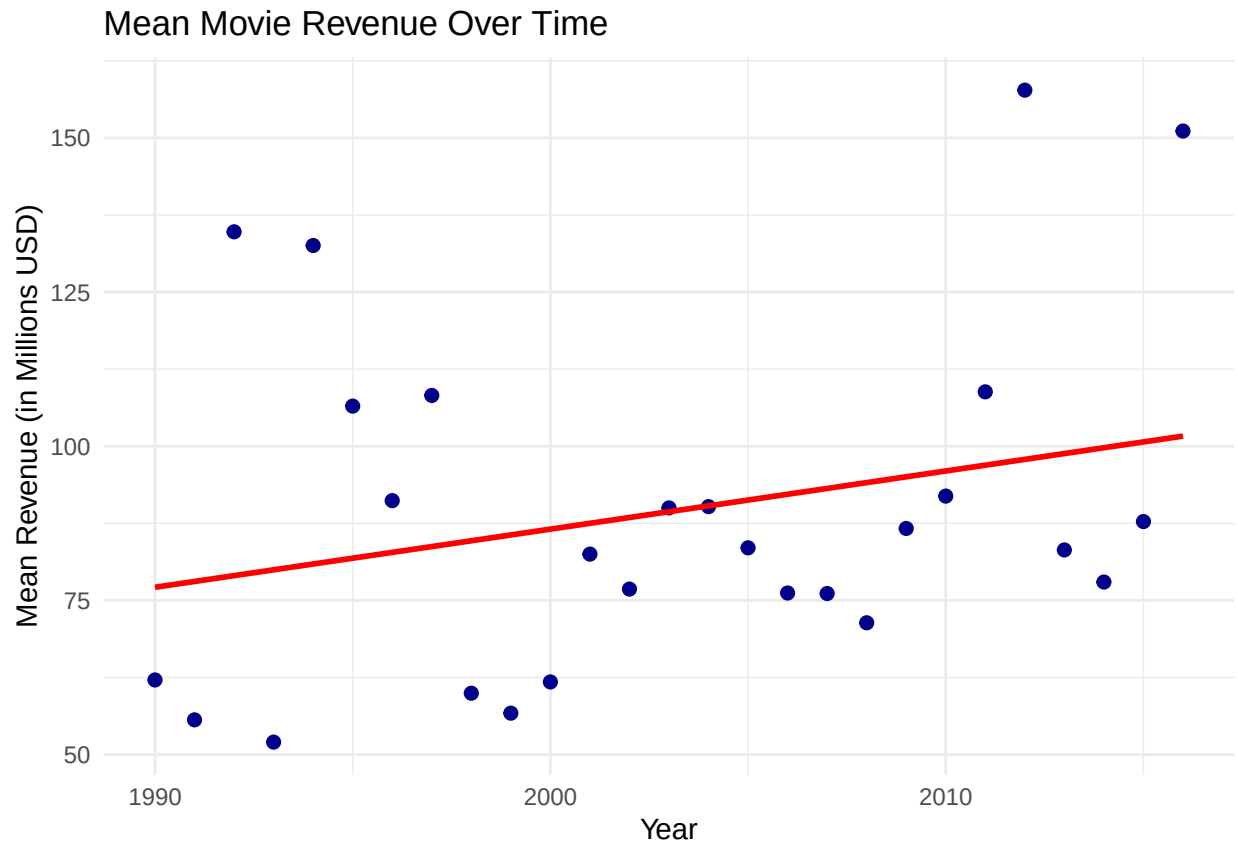
- Create a scatter plot with year on the x axis and mean of revenue within that year on the y-axis. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Is revenue of movies increasing or decreasing over time? **[3 points]**
- Recreate the scatter plot with year on the x axis and mean_revenue on the y-axis, but now add bars around each point, indicating the 95% confidence interval. **[3 points]**
- If you look closely at the resulting plot in b.), you will probably see that the confidence intervals in some years are wider than in other years. What are some possible reasons that a confidence interval in a given year can be wide? Can you verify this using your data? **[4 points]**

step a

```
library(ggplot2)
library(dplyr)

movies %>%
  group_by(release_year) %>%
  summarise(mean_revenue = mean(revenue_mil, na.rm = TRUE)) %>%
  ggplot(aes(x = release_year, y = mean_revenue)) +
  geom_point(color = "darkblue", size = 2) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(title = "Mean Movie Revenue Over Time", x = "Year",
  y = "Mean Revenue (in Millions USD)") +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Your Answer: The scatter plot displays the mean movie revenue (in millions of dollars) for each year present in the dataset. Based on the scatter plot, the revenue of movies is increasing over time. The data points and the linear trend line show a clear upward trajectory across the years, indicating that average movie revenues have grown higher in more recent years.

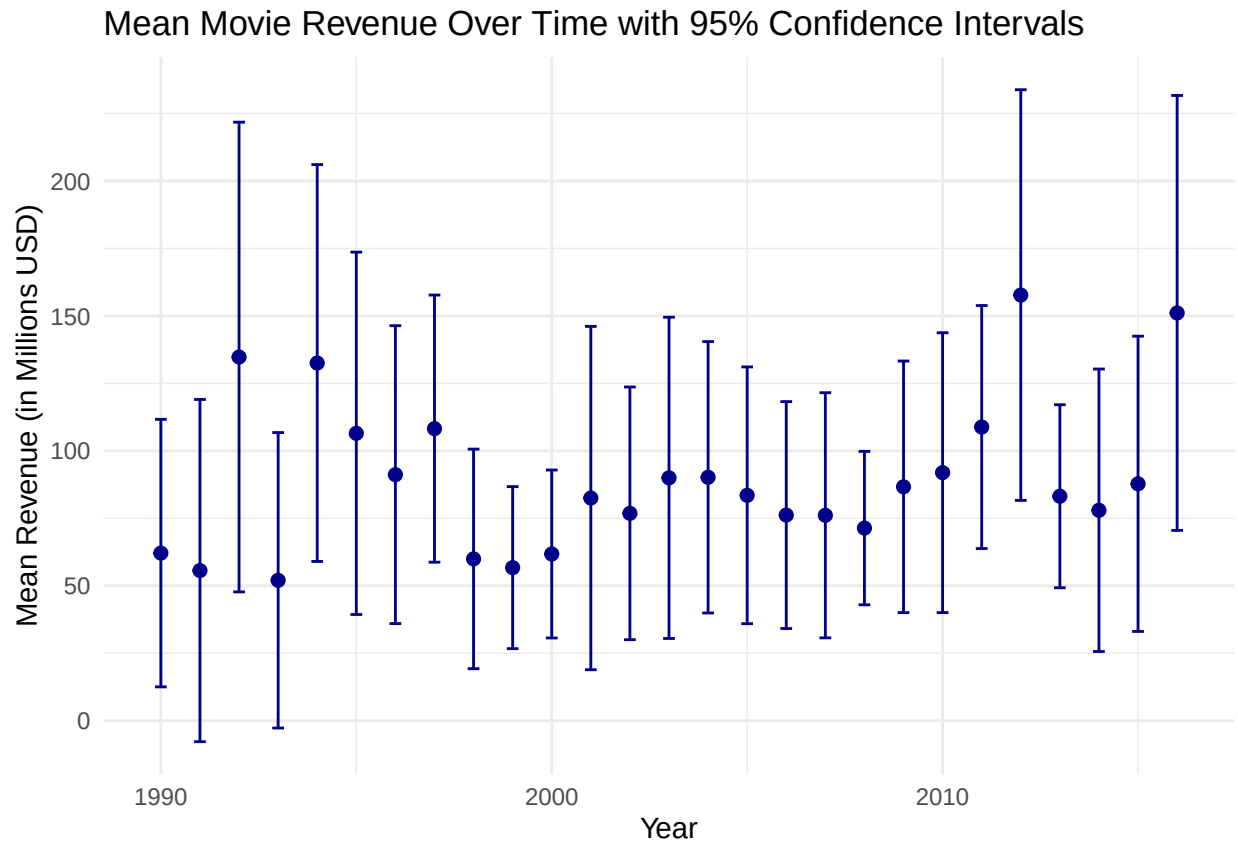
step b

```
library(dplyr)
library(ggplot2)

yearly_stats <- movies %>%
  group_by(release_year) %>%
  summarise(
    mean_revenue = mean(revenue_mil, na.rm = TRUE),
    sd_revenue = sd(revenue_mil, na.rm = TRUE),
    n = n(),
    se = sd_revenue / sqrt(n),
    ci_lower = mean_revenue - 1.96 * se,
    ci_upper = mean_revenue + 1.96 * se
  )

ggplot(yearly_stats, aes(x = release_year, y = mean_revenue)) +
  geom_point(color = "darkblue", size = 2) +
  geom_errorbar(aes(ymin = ci_lower, ymax = ci_upper), width = 0.3, color = "darkblue") +
```

```
labs(
  title = "Mean Movie Revenue Over Time with 95% Confidence Intervals",
  x = "Year",
  y = "Mean Revenue (in Millions USD)"
) +
theme_minimal()
```



Your Answer:

The scatter plot has been recreated to plot the mean movie revenue (in millions USD) against the release year, with error bars representing the 95% confidence intervals for each year's sample mean. The vertical error bars illustrate the precision and variability of the mean revenue estimates over time, where wider intervals reflect greater variance or smaller sample sizes in specific years.

step c

```
library(dplyr)

yearly_checks <- movies %>%
  group_by(release_year) %>%
  summarise(
    n_movies = n(),
    sd_revenue = sd(revenue_mil, na.rm = TRUE),
    se_revenue = sd_revenue / sqrt(n_movies),
    ci_width = 2 * 1.96 * se_revenue
  ) %>%
```

```
arrange(desc(ci_width))

head(yearly_checks)
```

```
## # A tibble: 6 x 5
##   release_year n_movies sd_revenue se_revenue ci_width
##         <dbl>   <int>    <dbl>    <dbl>    <dbl>
## 1      1992      10    140.     44.4    174.
## 2      2016     32    233.     41.1    161.
## 3      2012     45    260.     38.8    152.
## 4      1994     13    135.     37.5    147.
## 5      1995     17    141.     34.3    134.
## 6      2001     32    184.     32.5    127.
```

The width of a 95% confidence interval for the population mean depends on the standard error, which is calculated as $SE = s/\text{square root of } n$. There are two primary statistical reasons why a confidence interval in a given year can be particularly wide:

1. A smaller number of observed movies in a specific year decreases the denominator. So, less observations results in a wider confidence interval.
2. Higher variability or the presence of extreme box-office outliers in a given year increases the sample standard deviation.

Looking closely at the number per year, you can see that the wide intervals in years like 2012 and 2016 come down to two main things: the sample size (n) and how spread out the revenue numbers are. In those specific years, there weren't as many movies recorded in the dataset, which automatically makes the standard error bigger. On top of that, a few huge blockbuster movies made way more than the rest, which pulled up the standard deviation and made the confidence intervals even wider.